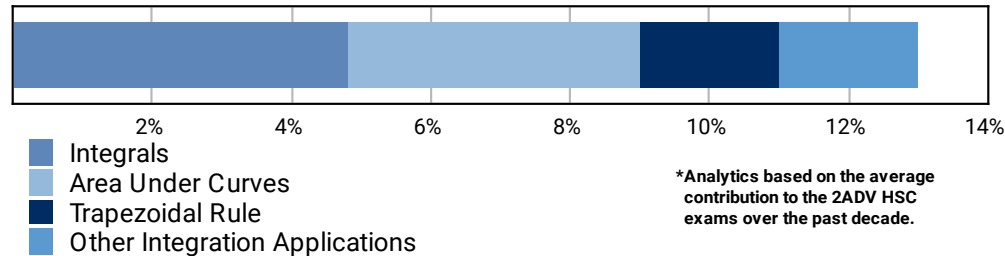


ADV: Calculus (Adv), C4 Integration (Adv), Integration (Y12)
L&E Integration

Teacher: Denis Vlismas

Exam Equivalent Time: 72 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C4 Integration



- We note 2019 13c(ii) surprisingly produced a sub-50% mean mark and deserves attention.

HISTORICAL CONTRIBUTION

- *C4 Integration* has contributed a substantial 13.0% to past Advanced exams, on average over the past decade.
- This topic has been split into four sub-topics for analysis purposes: 1-*Integrals* (4.8%), 2-*Areas Under Curves* (4.2%), 3-*Trapezoidal Rule* (2.0%) and 4-*Other Integration Examples* (2.0%).
- This analysis looks at *Integrals*.

HSC ANALYSIS - What to expect and common pitfalls

- *Integrals* (4.8%) is made up of standard integration (1.2%), log and exponential integration (2.4%) and trig integration (1.2%).
- Dedicated questions testing log and exponential integration have consistently appeared in past HSC exams - absent in just 2 exams of the past decade and including a record 3 separate questions in the 2020 exam.
- Questions requiring students to differentiate then integrate have easily been the worst answered, causing surprising problems in 2020 and 2019 - an important revision area.
- Integration of a fraction that results in a logarithm is a 2-3 mark question type that has appeared 5 times in the last decade, but just once in the last 5 years (2020 Adv 17).

Questions

1. Calculus, 2ADV C4 2014 HSC 4 MC

Which expression is equal to $\int e^{2x} dx$?

- (A) $e^{2x} + C$
- (B) $2e^{2x} + C$
- (C) $\frac{e^{2x}}{2} + C$
- (D) $\frac{e^{2x+1}}{2x+1} + C$

2. Calculus, 2ADV C4 2020 HSC 4 MC

What is $\int e + e^{3x} dx$?

- A. $ex + 3e^{3x} + c$
- B. $ex + \frac{1}{3}e^{3x} + c$
- C. $e + 3e^{3x} + c$
- D. $e + \frac{1}{3}e^{3x} + c$

3. Calculus, 2ADV C4 2012 HSC 9 MC

What is the value of $\int_1^4 \frac{1}{3x} dx$?

- (A) $\frac{1}{3} \ln 3$
- (B) $\frac{1}{3} \ln 4$
- (C) $\ln 9$
- (D) $\ln 12$

4. Calculus, 2ADV C4 2013 HSC 11e

Find $\int e^{4x+1} dx$ (2 marks)

5. Calculus, 2ADV C4 2004 HSC 3bi

Evaluate $\int_1^2 e^{3x} dx$. (2 marks)

6. Calculus, 2ADV C4 2006 HSC 2bi

Find $\int 1 + e^{7x} dx$. (2 marks)

7. Calculus, 2ADV C4 2015 HSC 11h

Find $\int \frac{x}{x^2 - 3} dx$. (2 marks)

8. Calculus, 2ADV C4 2018 HSC 11e

Evaluate $\int_0^3 e^{5x} dx$. (2 marks)

9. Calculus, 2ADV C4 2016 HSC 12d

i. Differentiate $y = xe^{3x}$. (1 mark)

ii. Hence find the exact value of $\int_0^2 e^{3x}(3 + 9x) dx$. (2 marks)

10. Calculus, 2ADV C4 2004 HSC 3bii

Find $\int \frac{x}{x^2 - 3} dx$. (2 marks)

11. Calculus, 2ADV C4 2006 HSC 2bii

Evaluate $\int_0^3 \frac{8x}{1 + x^2} dx$. (3 marks)

12. Calculus, 2ADV C4 2019 MET1-N 3

Evaluate $\int_0^1 e^x - e^{-x} dx$. (2 marks)

13. Calculus, 2ADV C4 2020 HSC 17

Find $\int \frac{x}{4+x^2} dx$. (2 marks)

14. Calculus, 2ADV C4 SM-Bank 1

If $m = \int_1^3 \frac{2}{x} dx$, express e^m in its simplest form. (2 marks)

15. Calculus, 2ADV C4 SM-Bank 3

If $f'(x) = \frac{x^2}{x^3+1}$ and $f(1) = \log_e 2$, find $f(x)$. (3 marks)

16. Calculus, 2ADV C4 2005 HSC 2ci

Find $\int \frac{6x^2}{x^3+1} dx$. (2 marks)

17. Calculus, 2ADV C4 2008 HSC 2ci

Find $\int \frac{dx}{x+5}$. (1 mark)

18. Calculus, 2ADV C4 2010 HSC 2dii

Find $\int \frac{x}{4+x^2} dx$. (2 marks)

19. Calculus, 2ADV C4 2011 HSC 4b

Evaluate $\int_e^{e^3} \frac{5}{x} dx$ (2 marks)

20. Calculus, 2ADV C4 2012 HSC 12b

Find $\int \frac{4x}{x^2+6} dx$. (2 marks)

21. Calculus, 2ADV C4 2013 HSC 11f

Evaluate $\int_0^1 \frac{x^2}{x^3+1} dx$ (3 marks)

22. Calculus, 2ADV C4 2019 HSC 13c

i. Differentiate $(\ln x)^2$. (2 marks)

ii. Hence, or otherwise, find $\int \frac{\ln x}{x} dx$. (1 mark)

23. Calculus, 2ADV C4 2020 HSC 18

a. Differentiate $e^{2x}(2x+1)$. (2 marks)

b. Hence, find $\int (x+1)e^{2x} dx$. (1 marks)

Worked Solutions

1. Calculus, 2ADV C4 2014 HSC 4 MC

$$\begin{aligned}\int e^{2x} dx \\&= \frac{1}{2}e^{2x} + C \\&\Rightarrow C\end{aligned}$$

2. Calculus, 2ADV C4 2020 HSC 4 MC

$$\begin{aligned}\int e + e^{3x} &= ex + \frac{1}{3}e^{3x} + c \\(\text{Note } e \text{ is a simple constant here}) \\&\Rightarrow B\end{aligned}$$

3. Calculus, 2ADV C4 2012 HSC 9 MC

$$\begin{aligned}\int_1^4 \frac{1}{3x} dx \\&= \frac{1}{3}[\ln x]_1^4 \\&= \frac{1}{3}[\ln 4 - \ln 1] \\&= \frac{1}{3}\ln 4 \\&\Rightarrow B\end{aligned}$$

TIP: Note that $\ln(1) = 0$, as $e^0 = 1$

4. Calculus, 2ADV C4 2013 HSC 11e

$$\int e^{4x+1} dx = \frac{1}{4}e^{4x+1} + C$$

5. Calculus, 2ADV C4 2004 HSC 3bi

$$\begin{aligned}\int_1^2 e^{3x} dx &= \frac{1}{3}[e^{3x}]_1^2 \\&= \frac{1}{3}(e^6 - e^3)\end{aligned}$$

6. Calculus, 2ADV C4 2006 HSC 2bi

$$\int 1 + e^{7x} dx = x + \frac{1}{7}e^{7x} + c$$

7. Calculus, 2ADV C4 2015 HSC 11h

$$\begin{aligned}\int \frac{x}{x^2 - 3} dx \\&= \frac{1}{2} \int \frac{2x}{x^2 - 3} dx \\&= \frac{1}{2} \ln(x^2 - 3) + C\end{aligned}$$

8. Calculus, 2ADV C4 2018 HSC 11e

$$\begin{aligned}\int_0^3 e^{5x} dx &= \left[\frac{1}{5}e^{5x} \right]_0^3 \\&= \frac{1}{5}e^{15} - \frac{1}{5}e^0 \\&= \frac{1}{5}(e^{15} - 1)\end{aligned}$$

9. Calculus, 2ADV C4 2016 HSC 12d

i. $y = xe^{3x}$

Using product rule:

$$\begin{aligned}\frac{dy}{dx} &= x \cdot 3e^{3x} + 1 \cdot e^{3x} \\&= e^{3x}(1 + 3x)\end{aligned}$$

ii. $\int_0^2 e^{3x}(3 + 9x) dx$

$$\begin{aligned}&= 3 \int_0^2 e^{3x}(1 + 3x) dx \\&= 3[xe^{3x}]_0^2 \\&= 3(2e^6 - 0) \\&= 6e^6\end{aligned}$$

10. Calculus, 2ADV C4 2004 HSC 3bii

$$\begin{aligned}\int \frac{x}{x^2 - 3} dx &= \frac{1}{2} \int \frac{2x}{x^2 - 3} dx \\ &= \frac{1}{2} \ln(x^2 - 3) + C\end{aligned}$$

11. Calculus, 2ADV C4 2006 HSC 2bii

$$\begin{aligned}\int_0^3 \frac{8x}{1+x^2} dx &= 4 \int_0^3 \frac{2x}{1+x^2} dx \\ &= 4 [\log_e(1+x^2)]_0^3 \\ &= 4 [\log_e(1+9) - \log_e(1+0)] \\ &= 4 [\log_e 10 - \log_e 1] \\ &= 4 \log_e 10\end{aligned}$$

12. Calculus, 2ADV C4 2019 MET1-N 3

$$\begin{aligned}\int_0^1 e^x - e^{-x} dx &= [e^x + e^{-x}]_0^1 \\ &= \left[e + \frac{1}{e} - (1 + 1) \right] \\ &= e + \frac{1}{e} - 2\end{aligned}$$

13. Calculus, 2ADV C4 2020 HSC 17

$$\begin{aligned}\int \frac{x}{4+x^2} dx &= \frac{1}{2} \int \frac{2x}{4+x^2} dx \\ &= \frac{1}{2} \log_e(4+x^2) + c\end{aligned}$$

14. Calculus, 2ADV C4 SM-Bank 1

$$\begin{aligned}\int_1^3 \frac{2}{x} dx &= [2 \log_e x]_1^3 \\ m &= 2 \log_e 3 - 2 \log_e 1 \\ &= 2 \log_e 3 \\ \therefore e^m &= e^{2 \log_e 3} \\ &= e^{\log_e 9} \\ &= 9\end{aligned}$$

15. Calculus, 2ADV C4 SM-Bank 3

$$\begin{aligned}f(x) &= \int f'(x) dx \\ &= \int \frac{x^2}{x^3 + 1} dx \\ &= \frac{1}{3} \int \frac{3x^2}{x^3 + 1} dx \\ &= \frac{1}{3} \log_e |(x^3 + 1)| + c\end{aligned}$$

Since $f(1) = \log_e 2$,

$$\begin{aligned}\frac{1}{3} \log_e 2 + c &= \log_e 2 \\ c &= \frac{2}{3} \log_e 2\end{aligned}$$

$$\therefore f(x) = \frac{1}{3} \log_e(x^3 + 1) + \frac{2}{3} \log_e 2$$

16. Calculus, 2ADV C4 2005 HSC 2ci

$$\begin{aligned}\int \frac{6x^2}{x^3 + 1} dx &= 2 \int \frac{3x^2}{x^3 + 1} dx \\ &= 2 \ln(x^3 + 1) + C\end{aligned}$$

17. Calculus, 2ADV C4 2008 HSC 2ci

$$\int \frac{dx}{x+5}$$

$$= \ln(x+5) + C$$

18. Calculus, 2ADV C4 2010 HSC 2dii

$$\int \frac{x}{4+x^2} dx$$

$$= \frac{1}{2} \int \frac{2x}{4+x^2} dx$$

$$= \frac{1}{2} \ln(4+x^2) + C$$

IMPORTANT: Minimise errors by adjusting the integral to fit the form $\frac{f'(x)}{f(x)}$ before integrating.

19. Calculus, 2ADV C4 2011 HSC 4b

$$\int_e^{e^3} \frac{5}{x} dx$$

$$= 5 \int_e^{e^3} \frac{1}{x} dx$$

$$= 5[\ln x]_e^{e^3}$$

$$= 5(\ln e^3 - \ln e)$$

$$= 5(3 - 1)$$

$$= 10$$

MARKER'S COMMENT: Most common error was $\ln(5x)$. Minimize errors by getting the integral in the form of $\frac{f'(x)}{f(x)}$ before integrating.

20. Calculus, 2ADV C4 2012 HSC 12b

$$\int \frac{4x}{x^2+6} dx$$

$$= 2 \int \frac{2x}{x^2+6} dx$$

$$= 2 \ln(x^2+6) + C$$

21. Calculus, 2ADV C4 2013 HSC 11f

$$\int_0^1 \frac{x^2}{x^3+1} dx$$

$$= \frac{1}{3} \int_0^1 \frac{3x^2}{x^3+1} dx$$

$$= \frac{1}{3} [\ln(x^3+1)]_0^1$$

$$= \frac{1}{3} (\ln 2 - \ln 1)$$

$$= \frac{1}{3} \ln 2$$

TIP: Note that $\ln(1) = 0$ because $e^0 = 1$

22. Calculus, 2ADV C4 2019 HSC 13c

i. $y = (\ln x)^2$

$$\frac{dy}{dx} = 2 \cdot \frac{1}{x} \cdot \ln x$$

$$= \frac{2 \ln x}{x}$$

ii. $\int \frac{\ln x}{x} dx = \frac{1}{2} \int \frac{2 \ln x}{x} dx$

$$= \frac{1}{2} (\ln x)^2 + C$$

♦ Mean mark part (ii) 49%.

a. $y = e^{2x}(2x + 1)$

$$\frac{dy}{dx} = 2x^{2x}(2x + 1) + 2e^{2x}$$

$$= 2e^{2x}(2x + 2)$$

$$= 4e^{2x}(x + 1)$$

♦ Mean mark part (b) 40%.

b. $\int (x + 1)e^{2x} dx = \frac{1}{4} \int 4e^{2x}(x + 1)$

$$= \frac{1}{4} e^{2x}(2x + 1) + c$$